

THE 8 DOLLAR FILTER HABIT THAT SAVES A 6,000 DOLLAR FURNACE

A clogged filter can overheat the heat exchanger in a gas furnace. This is the once a month habit, the size lookup, and the carbon monoxide safety line that matters more than any of it.

WHAT IT IS

An 8 dollar filter that clogs can overheat the heat exchanger in a gas furnace that costs about 6,000 dollars to replace. The fix is a habit, not a tool. This sheet covers gas furnaces. On a heat pump or electric furnace the filter habit still protects airflow, but the heat exchanger danger belongs to gas.

The filter sits upstream of the blower, in a return grille, a duct, or a slot in the furnace. When it clogs, less air moves across the hot metal, and the same heat has to go into less air. The air, and the metal, come out hotter.

THE MONTHLY CHECK

- ☐ **Once a month, pull the filter and hold it up to a light.** If you cannot see light through most of it, change it.
- ☐ **Change it at least every 3 months** even if it looks fine.
- ☐ Change it sooner in a high fouling home. Pets, heavy dust, and remodeling work load a filter up fast.
- ☐ Put a reminder on your phone for the first of every month.

Weak air at the vents, rooms that heat slowly, or a furnace that shuts off early all mean the filter may be loaded. Change it before the calendar says.

THE MONEY

8 dollars	Rough cost of one common one inch pleated filter bought in a pack.
about 32 dollars	Four filters a year, the cost of the whole habit.
about 4,800 dollars	Average cost to replace a gas furnace. A 6,000 dollar bill sits well inside the range.
480 to 640 dollars	What the filter habit adds up to over 15 to 20 years, about a tenth of a 6,000 dollar furnace.

WHAT A CLEAN FILTER CAN AND CANNOT DO

A clean filter removes one known cause of overheating for almost no money. It does not guarantee a 20 year furnace, it tells you nothing about a cracked heat exchanger or carbon monoxide, and it cannot keep the furnace running in a blackout. Those are on pages 2 and 4.

The wrong replacement filter can cause the same overheating, and it is easy to buy one thinking it is an upgrade. Page 3 covers how to choose.

Scoreboard 8 dollar habit vs 6,000 dollar repair	
THE HABIT About 32 dollars a year, five minutes a month, removes a known cause of overheating.	THE NEGLECT A restricted filter can trip the limit switch and damage the heat exchanger, the part that can cost thousands to replace.

WHAT HAPPENS INSIDE

A gas furnace burns fuel in burners, and the flames heat a metal heat exchanger. That metal keeps the combustion gases apart from your house air, and the exhaust leaves through the vent. A blower pushes house air across the outside of the metal, and that air carries the heat into your ducts. A single stage furnace fires at one fixed rate while its burners run.

THE ARITHMETIC OF A HOT FURNACE

Technicians call the difference between the air going in and the air coming out the temperature rise.

heat = airflow x temperature rise (F) x about 1.08

The answer is the heat, in British thermal units per hour, that the air carries away. Worked example: a furnace burns 80,000 British thermal units of gas per hour, and about 80 percent reaches your house air. That is 64,000 units of heat per hour going into the air.

These airflow drops are made up to show the math. A real clogged filter rarely cuts airflow this far. The direction still holds: less air carrying the same heat comes out hotter.

AIRFLOW VS TEMPERATURE RISE

Airflow (cubic feet per minute)	Temperature rise	vs a 45 to 75 plate
1,200	about 49 degrees	fine
900	about 66 degrees	getting high
700	about 85 degrees	ten degrees over the top

Based on 64,000 units of heat per hour from the worked example.

THE RATING PLATE AND THE LIMIT SWITCH

- Every gas furnace has a rating plate, a label listing the temperature rise it is built for. A furnace training site says typical ranges are 45 to 75 degrees, or 50 to 80.
- The limit switch is a safety sensor. If the heat exchanger area gets too hot, it shuts the burners off, then lets them relight once things cool.
- Run near the top of the range and the furnace can start cycling on that limit. The same site says the limit is not built to click on and off all winter, and eventually it fails.

A Goodman and Amana installation manual says restricted airflow can cause excessive temperature rise, limit switch tripping, and heat exchanger failure, and lists the filter right alongside the ducts and grilles.

Want your own number? A technician measures the rise with a probe thermometer in the return and another in the supply, then compares the difference with the plate.

WHAT A DIRTY FILTER REALLY DOES

10 homes

Lawrence Berkeley National Laboratory monitored ten California homes for at least a year each.

under 1%

In most of them, filter loading changed total energy use by under 1 percent.

Conventional blowers generally lost airflow as the filter loaded with dust. Blowers built to hold airflow steady did something sneakier: they sped up and used more electricity to keep the air moving. The rooms feel fine while the motor quietly works harder. Once the motor hits top speed, the airflow finally starts to fall.

One home fouled its filter at a high rate. The researchers concluded that changing filters more often in high fouling homes is essential.

No source counts furnaces that died early from a dirty filter. What the evidence supports is a chain: a loaded filter can cut airflow, less airflow raises the temperature rise, and a rise above the plate range can trip the limit switch and damage parts. How fast depends on the blower and the house.

OTHER CAUSES A FILTER CANNOT FIX

- Low airflow, a wrong orifice size, or overfiring can crack a newer heat exchanger.
- A dirty blower wheel can choke airflow too. Cleaning it takes a technician, a good reason to book a yearly tune up.
- Lennox says a gas furnace averages 15 to 20 years. Energy Star agrees a dirty filter slows airflow and the dust buildup can mean expensive repairs or early failure.

CHOOSING THE FILTER

Filters carry a rating called Merv, short for minimum efficiency reporting value. A higher number catches smaller particles, but higher is not automatically better, because higher ratings often resist airflow more.

5 to 9 dollars

Rough price of a common one inch pleated Merv 8 filter bought in a pack. Prices vary by store and size.

7% and 11%

Airflow drop in two systems in a Texas home, tested by researcher Brent Stephens, when a low rated filter was swapped for a high rated one.

up to 60%

Simulated cut in airflow and blower power from a heavily loaded Merv 16 on a conventional blower. Berkeley also saw noise complaints and air bypassing the filter.

THE RULE

- ☐ Use the filter size and type your furnace manual specifies.
- ☐ No manual? Match the size and rating of the filter in there now.
- ☐ Want a higher rating? Ask a technician to check your airflow first.
- ☐ A Daikin manual warns against very restrictive one inch filters, and says a deep pleated filter rated Merv 8 or better often restricts airflow less than a standard one inch and needs replacing less often.
- ☐ Write the size and rating on the furnace door with a marker so nobody guesses at the store.

Scoreboard

Higher Merv is not a free upgrade

WRONG MOVE

Dropping in the most restrictive high rated one inch filter can cut airflow and cause the same overheating you are trying to prevent.

RIGHT MOVE

Match the manual, or ask a technician before going higher. A deep pleated Merv 8 or better is the upgrade to ask about if you have room for a deep cabinet.

CHANGING THE FILTER

- 1 If you smell gas, leave the house right away without touching any switches, and call your gas company or emergency services from outside.
- 2 Turn the thermostat off and let the blower stop.
- 3 Switch off furnace power, at the switch near the furnace or at the breaker, before you open any panel.
- 4 Find the filter slot: near the blower, in an external cabinet, or behind a large return grille.
- 5 Slide the old filter out and hold it up to the light. No light through it means it is overdue.
- 6 Read the size on the frame, something like 16 by 25 by 1. Snap a photo of the label before you toss it.
- 7 Check the arrow on the new filter. It points the way the air moves, toward the furnace and blower. Backward restricts airflow.
- 8 Write the date on the frame, slide the new filter in snug, and close the door or grille all the way.
- 9 Switch furnace power back on, then turn the thermostat on.

DO NOT DO THESE

- Never run the furnace with the filter slot empty. A Daikin manual says damage from running without filters is not covered by the warranty.
- Do not add a second filter at the furnace rack if one already lives at the return grille, without asking a technician.
- Never bypass or reset a limit switch to keep a furnace running.
- If the furnace keeps starting and shutting off after short runs, even with a fresh filter, turn the heat off and call a technician.

THE CARBON MONOXIDE RISK

A normal temperature rise does not prove the heat exchanger is intact or the combustion is safe. Changing the filter tells you nothing about a cracked heat exchanger. Those checks take a professional inspection.

- ☐ The Consumer Product Safety Commission calls fuel burning heating systems a potential source of carbon monoxide, an odorless, colorless, poisonous gas. It reports more than 200 accidental carbon monoxide deaths a year, on average, from consumer products other than cars.
- ☐ **Get a professional inspection of your heating system every year.**
- ☐ **Put carbon monoxide alarms on every level and outside sleeping areas.** Choose battery power or battery backup, test them, and replace them when they expire.
- ☐ **If an alarm sounds, or you feel headache, dizziness, weakness, nausea, or confusion,** get outside to fresh air and call nine one one or the fire department. Do not go back in until responders say it is safe.

IT CANNOT FIX A BLACKOUT

When the power goes out, a typical gas furnace will not produce heat, even with gas flowing and a spotless filter. The gas valve needs 24 volt power from the control board, and the blower and ignition need electricity too. Backup power that feeds the whole furnace is a job for an electrician. Run portable generators outdoors, at least 20 feet from the house, and never connect one through a wall outlet.

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